Analytica Chimica Acta 748 (2012) 58–66

Contents lists available at SciVerse ScienceDirect

Analytica Chimica Acta

journal homepage: www.elsevier.com/locate/aca

Use of temperature dependent Raman spectra to improve accuracy for analysis of complex oil-based samples: Lube base oils and adulterated olive oils

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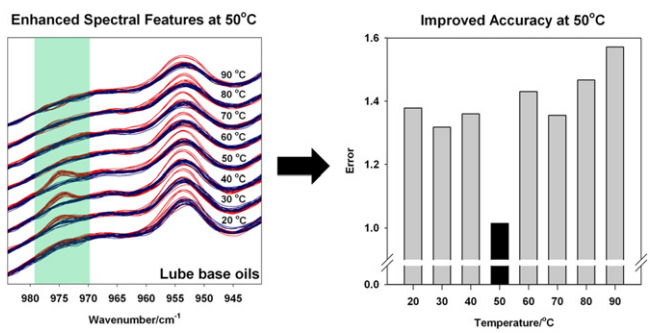
HIGHLIGHTS

- Temperature dependent Raman spectral variation was effectively used to improve accuracy for analysis of complex oil-based samples.
- Enhanced spectral features of lube base oils occurred at 50 °C described the variation of viscosity more accurately.
- The temperature-induced spectral variation around 80–90 °C improved the authentication of adulterated olive oils.

GRAPHICAL ABSTRACT



ARTICLE INFO

Article history:
Received 23 April 2012
Received in revised form 31 July 2012
Accepted 9 August 2012
Available online 23 August 2012

Keywords:
Raman spectral selectivity
Temperature-sensitive molecular vibration
Generalized two-dimensional correlation analysis
Lube base oil
Viscosity
Adulterated olive oil

ABSTRACT

A simple and effective strategy to improve accuracy for Raman spectroscopic analysis of complex mixture samples by probing a measurement temperature yielding enhanced spectral selectivity has been demonstrated. For the evaluation, the determination of Kinematic Viscosity at 40 °C (KV@40) of lube base oil (LBO) samples was initially attempted. Partial least squares (PLS) was used to determine the KV@40 using Raman spectra of the samples collected at 8 different temperatures from 20 to 90 °C with 10 °C increments. Interestingly, the distinct temperature-induced spectral variation among the samples occurred at 50 °C, thereby resulting in the improved accuracy for determination of KV@40. Two-dimensional (2D) correlation analysis was also performed to find an additional supportive rationale for the improved accuracy. The strategy was further evaluated for the identification of soybean oil-adulterated olive oils using linear discriminant analysis (LDA). Similarly, the discrimination accuracy was improved around 80–90 °C due to the enhanced spectral selectivity between olive and soybean oils. In overall, these two results successfully demonstrate analytical effectiveness of the strategy.

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1. Introduction

Near-infrared (NIR) [1,2] and Raman spectroscopy [3] has been widely utilized for quantitative analysis of samples in diverse fields such as agriculture [4–6], pharmaceutical [6–9] and petroleum refining [10,11], since these enable fast and non-destructive analysis. The spectral features obtained from these samples often overlap due to the complexity in chemical composition. Therefore, multivariate methods such as partial least squares (PLS) [12,13] are necessarily employed to establish an empirical correlation between the spectral feature and the corresponding property (or composition). To improve the measurement accuracy, several analytical approaches are available such as the collection of high resolution spectra and proper use of chemometric methods without overfitting. Nonetheless, the enhancement of spectral selectivity (or spectroscopic discriminability) among samples could be a more

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http://dx.doi.org/10.1016/j.aca.2012.08.028

basic and effective approach. The better spectral selectivity implies that the spectral features among the samples are more discernible each other. Alternatively the employment of separation methods may guarantee the selectivity of measurement; however, it also has limitations when fast non-destructive analysis without any sample pre-treatments is in a great demand.

Recently we demonstrated the improvement of accuracy for determination of polyethylene (PE) density by collecting Raman spectra of the samples at 70 °C [14]. The pre-melt behavior of the PE samples occurred at this temperature enhanced Raman spectral selectivity among the samples, thereby providing more accurate determination of PE density. The objective of the work in this publication is to improve accuracy for Raman spectroscopic analysis of complex mixture samples by investigating a strategy of maximizing their spectral selectivity. When the temperature of a mixture sample varies, the temperature-induced spectral variations of individual components as well as their inter-molecular interaction should be diverse. Therefore, at a certain temperature, the enhanced spectral selectivity among samples would be available and help to improve accuracy of both quantitative and qualitative analysis.

First, lube base oil (LBO) was chosen as a complex mixture sample and determination of Kinematic Viscosity at 40 °C (KV@40), a bulk property individually contributed from each component in LBO, was attempted. As known, LBO is composed of widely different hydrocarbons, such as long-chain paraffins, branched paraffins and polynuclear aromatic compounds. To examine the dependence of accuracy on the measurement temperature, partial least squares (PLS) regression [12,13] was used to determine KV@40 using Raman spectra of LBO samples collected at 8 different temperatures from 20 to 90 °C with 10 °C intervals and an optimal temperature yielding a significant improvement in accuracy was searched. To alternatively investigate the spectral variation of the samples at each temperature, two-dimensional (2D) correlation analysis [15–18] was used.

Development of fast and simple analytical methods for the authentication of olive oil is now in demand [19–22,24] since adulteration of olive oils with cheaper frying oils has increased especially in international trading. So, in second, the variation in identification accuracy for adulterated olive oils according to the temperature of Raman measurement was investigated. Using Raman spectra of soybean oil-adulterated olive oils collected at the same 8 different temperatures, linear discriminant analysis (LDA) [25–27] combined with principal component analysis (PCA) [12,13] was performed and an optimal temperature providing the improved identification was probed. The loadings employed for the measurements at each temperature were compared to investigate the difference in the weighted spectral features in each model.

2. Experimental

2.1. Samples

Thirty different LBO samples produced from diverse international manufacturers were used in this study. The samples were carefully prepared to represent a wide variation of KV@40. KV@40 of the samples was determined using the method of ASTM D445. The colors of samples varied from transparent to light yellow. Although the samples were non-volatile, these were tightly sealed and stored in a refrigerator.

Extra virgin olive and soybean oils used in this study were purchased at a local market. Ten separate olive oils were produced from two different manufactures (5 different bottles of the same olive oils from each manufacturer). Also, 10 soybean oils were produced from two manufacturers. Ten adulterated olive oils were prepared by randomly blending olive and soybean oils without duplicated mixing. The concentration of soybean oil was 5 vol% for all adulterated olive oils.

2.2. Raman spectral collection and data processing

Raman spectra were collected using the wide area illumination (WAI) scheme (excitation: 785 nm) as employed in a previous study [28]. Both samples were transferred into a circular glass vial (diameter: 9.8 mm). Then, the vial contained with a sample was positioned in a metal block capable of controlling the temperature by feedback control to a precision of ±0.1 °C. The power of the laser beam at the glass vial was 100 mW. Raman spectra of both samples were collected at 8 different temperatures from 20 to 90 °C, with 10 °C intervals. A laser was exposed over 40 s for each spectral collection. Triplicate spectra collected for each LBO sample and 5 replicate spectra were collected for each adulterated olive oil. At each replicate spectral collection, new fresh samples were used. The resolution of collected spectra was 4 cm^{-1}.

All spectral pre-treatments as well as multivariate analyses such as baseline correction, normalization, PLS, 2D correlation analysis, PCA and LDA were performed using MATLAB Version 7.0 (Math Works Inc., MA, USA).

3. Results and discussion

3.1. Raman spectral features of LBO samples

Fig. 1(a) shows normalized Raman spectra (1217–690 cm^{-1} range) of 4 LBO samples with different KV@40 s collected at 20 °C. The baselines of raw spectra were linearly corrected at 1217, 1115, 989 and 690 cm^{-1}, and area under the 1217–690 cm^{-1} range was calculated. Then, each baseline-corrected spectrum was divided by the corresponding area for normalization. Among these four samples, two samples correspond to the lowest (16.43 cSt, red) and highest (42.15 cSt, blue) KV@40 s in the dataset and the other two samples have medium KV@40 s as indicated in Fig. 1. The relevant band assignments can be found in other publications [29–31]. For detail examination of the spectral features according to the variation of KV@40, the 1082–1058, 962–945 and 900–790 cm^{-1} ranges were individually magnified in Fig. 1(b)–(d), respectively.

As shown in Fig. 1(b), the shift of band position as well as change of bandwidth is observed upon the variation of KV@40. The intensity of 953 cm^{-1} band (C—C skeletal stretching) decreases with the increase of KV@40 as shown in Fig. 1(c). In addition, when KV@40 becomes higher, the band shape (bottom 2 spectra) changes to an overlapped feature with two or more components. The bands in the 900–790 cm^{-1} range (Fig. 1(d)) show complex spectral variations according to the change of KV@40 such as the increase/decrease of band intensity and slight band shift. Overall, the systematic spectral variation according to the change of KV@40 is difficult to observe in these figures since KV@40 is a bulk physical property resulting from the contributions from individual components as well as interactions among them. However, the variation of KV@40 is related with the compositional changes of samples. Therefore, Raman spectra reflecting the change in composition as well as inter-molecular interactions are useful to describe the variation of KV@40.

Fig. 2 shows normalized Raman spectra of a LBO sample (KV@40: 18.6 cSt) collected at 8 different temperatures in the 1217–690 cm^{-1} range. As shown, upon the temperature perturbation, the corresponding Raman spectral features vary substantially. Raman intensity of the bands centered at 1150 (skeletal stretching primarily in *iso*-alkanes), 1076 (skeletal stretching of alkanes and alkenes) and 1063 cm^{-1} (stretching primarily in *n*-alkanes) decreases with the temperature elevation. Since density of the

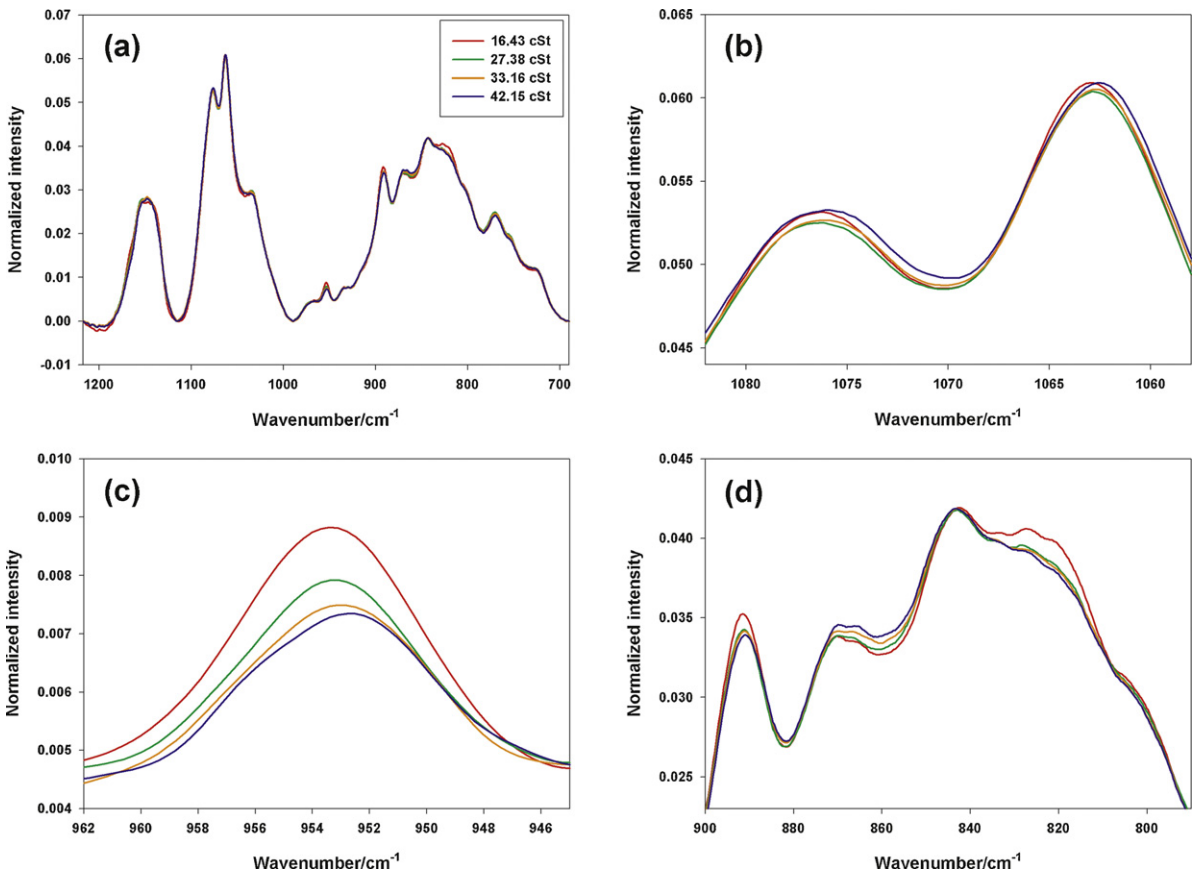

Fig. 1. Normalized Raman spectra (1217–690 cm^{-1} range) of 4 LBO samples with different KV@40 s collected at 20 °C (a). Among these four samples, two samples correspond to the lowest (16.43 cSt, red) and highest (42.15 cSt, blue) KV@40 s and the other two samples have medium KV@40 s. KV@40 s of the samples are designated in the figure. For detail examination of the spectral features according to the variation of KV@40, the 1082–1058 (b), 962–945 (c) and 900–790 cm^{-1} (d) ranges are individually highlighted. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

sample decreases at elevated temperature, an absolute number of molecules present in a given laser illumination volume decreases and so the lowered Raman intensity is resultant. During the temperature variation, the position of 1063 cm^{-1} band remains unchanged; while, the position as well as shape of 1076 cm^{-1} band change minutely, although the changes are hardly recognizable in the figure. Reversely, in the 880–790 cm^{-1} range, the intensity of bands at 870, 844 (alkylated saturated ring breathing) and 831 cm^{-1} (symmetric C—C—C stretch) increases with the temperature elevation. In addition, the detail band shapes in this range are different each other. The observation indicates the occurrence of complex and diverse temperature-induced spectral variations of individual components.

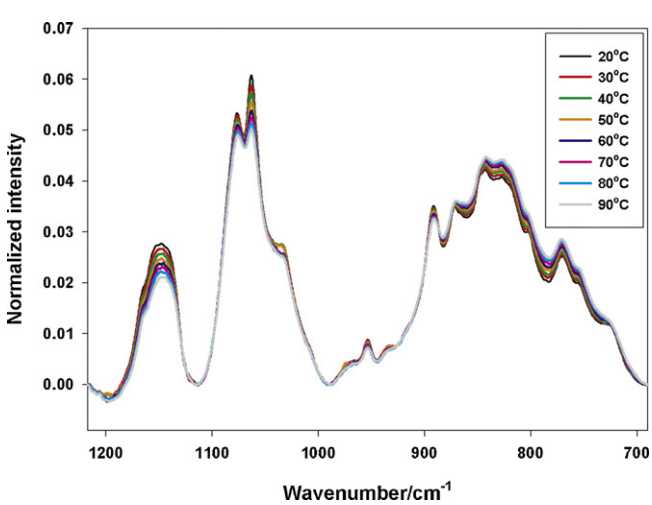

Fig. 2. Normalized Raman spectra of a LBO sample (KV@40: 18.6 cSt) collected at 8 different temperatures in the 1217–690 cm^{-1} range. Each color indicates the temperature of measurement. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

3.2. Determination of KV@40 at each temperature using PLS

Partial least squares (PLS) [12,13] was used to determine KV@40 of LBO samples using the datasets collected at 8 different temperatures. The 1217–690 cm^{-1} range shown in Figs. 1 and 2 was used for PLS. The errors for the determination of KV@40 were obtained by full cross-validation. For this purpose, 23 and 7 samples were randomly assigned to the calibration and validation sets, respectively. The standard error of calibration (SEC) and standard error of prediction (SEP) were obtained at all possible combinations of the sample assignment. Finally, mean standard error of calibration (MSEC) and mean standard error of prediction (MSEP) were calculated by averaging the individual SEC and SEP values.

Fig. 3 shows the variation of MSEP at 8 different sample measurement temperatures. In all cases, 7 factors were used for PLS. As shown, the MSEP abruptly drops at the 50 °C measurement. The improvement of accuracy at this temperature is statistically

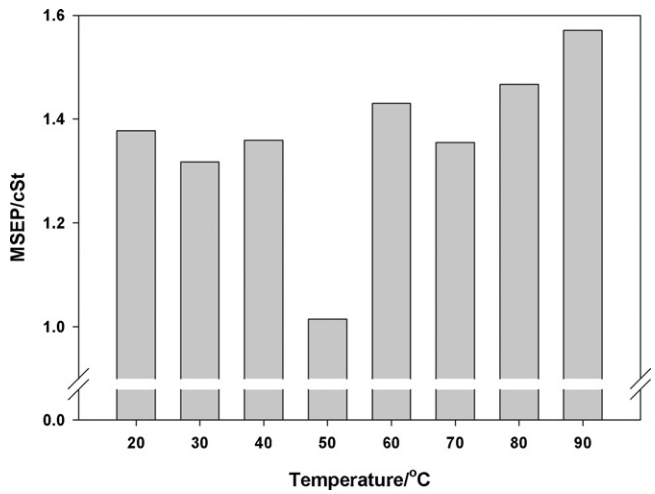

Fig. 3. Mean standard error of prediction (MSEP) for the determination of KV@40 using Raman spectra of LBO samples collected at 8 different measurement temperatures. In all cases, 7 factors were used for PLS.

significant based on a *t*-test at 95% confidence level. The result indirectly supports that Raman spectral features of LBO samples acquired at 50 °C provide more descriptive information to follow the variation of KV@40 more accurately.

3.3. Varied Raman spectral features of LBOs induced by temperature elevation

To find a rationale for the improved accuracy, it is initially required to examine the spectral variations occurring at 50 °C. Fig. 4(a) shows Raman spectra of all LBO samples collected at 8 different temperatures in the 1049–1023 cm^{-1} range, where the distinct spectral variation is observed with the temperature elevation. Two colors (blue: 14 samples, red: 16 samples) are used for the better recognition of spectral variation. At 20 °C, the 1035 cm^{-1} bands from all samples are largely mixed with each other. From 30 °C, roughly two spectral groups starts to observe and the most distinct discrimination occurs at 50 °C (also refer to the inset in the figure). Moreover, at 50 °C, the shape of 1035 cm^{-1} band in the red-colored spectra is quite different such as the presence of two neighboring peaks. After 50 °C, the distinct bands at 50 °C suddenly disappear and both colored spectra are nearly mixed without indication of two spectral groups. The 1035 cm^{-1} band is assignable to C—C skeletal stretching [30,31], so it would expect that the backbone vibrations of long chain hydrocarbons in the samples are dissimilar at 50 °C.

Fig. 4(b) also shows Raman spectra highlighting the temperature-induced spectral variation in the 989–920 cm^{-1} range. The trend of spectral variation in this spectral range is similar to that observed in Fig. 4(a). Initially the 975 cm^{-1} band, corresponding to CH_2 wag in dialkyl ethylenes [30,31], is absent at 20 °C and starts to grow from 30 °C as shown in the red-colored spectra. The intensity of band is maximized at 50 °C for these 16 samples and suddenly the band disappears after 50 °C.

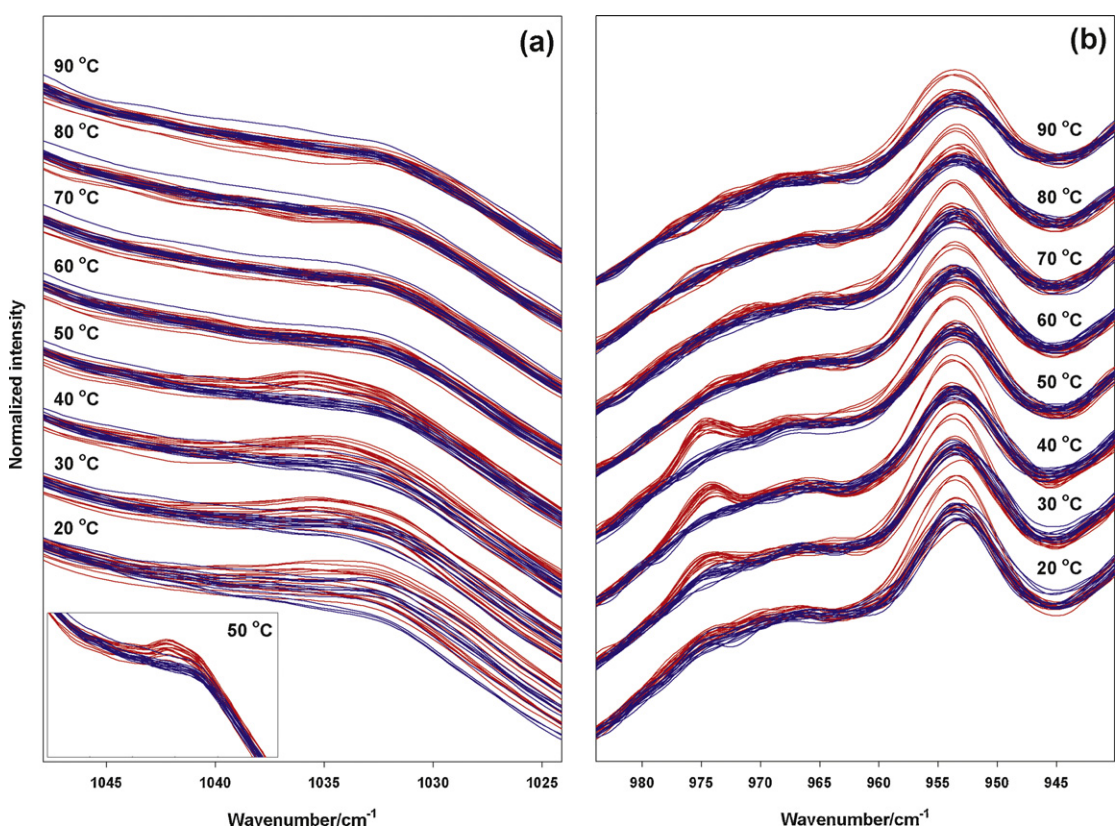

Fig. 4. Raman spectra of all LBO samples collected at 8 different temperatures in the 1049–1023 cm^{-1} (a) and 989–920 cm^{-1} (b) ranges. The spectra collected at each temperature are arbitrarily offset for the comparison. Two colors (blue: 14 samples, red: 16 samples) are used for the better recognition of spectral variation. The inset highlights the spectral features around 1035 cm^{-1} at 50 °C. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

The samples belong to each group were produced with different catalysts, therefore the resulting chemical compositions between two sample groups could be also different. The molecular vibrations of components in the samples of red group vary more sensitively at 50 °C, so the characteristic bands appear (refer to Fig. 4) as a result.

3.4. Examination of PLS loadings used at each measurement

The examination of PLS loadings could be beneficial to probe the origin of improved accuracy at 50 °C. The first loadings at each measurement temperature in the 1217–1050, 1050–860 and 860–690 cm^{-1} ranges are shown in Fig. 5(a)–(c), respectively. The features of first loadings in the 1217–1050 cm^{-1} range are generally similar each other (refer to Fig. 5(a)). However, in the 1050–860 cm^{-1} range, the loading features at 1035 and 974 cm^{-1} are substantially different especially at 40 and 50 °C as shown in Fig. 5(b). The weights on these two bands are strongest at the 50 °C measurement and the positions of the weights exactly match with those of the bands showing distinct spectral variation at 50 °C (refer to Fig. 4). In addition, the loading features at 50 °C in the 750–690 cm^{-1} range are uniquely different from those acquired at other temperatures as shown in Fig. 5(c). Based on the observation of the loadings, the temperature-induced spectral variation is effectively reflected in the PLS models to correlate with the variation of KV@40.

3.5. Two-dimensional (2D) correlation analysis

For the alternative examination of spectral variation, a generalized 2D correlation analysis [15–18] was employed. Generalized 2D correlation analysis commonly provides the qualitative evaluation of spectral variations when perturbation is applied to samples. In this study, the spectral datasets collected at each temperature were individually used for 2D correlation analysis. Therefore, the spectral variation present in each dataset can be regarded as a result of perturbation induced by the change of KV@40 at a given temperature.

Fig. 6(a) and (b) shows the asynchronous 2D correlation maps acquired from the measurements at 20 to 50 °C, respectively. The same 1217–690 cm^{-1} range was used for the analysis. The asynchronous correlation was primarily investigated since it could recognize minute spectral variation more effectively compared to the synchronous correlation analysis [15–17]. Red and blue colors indicate the positive and negative cross peaks, respectively. In the asynchronous 2D correlation map at 20 °C, many small positive and negative correlation peaks are present. It indicates that complex and minute spectral variations occur upon the change of KV@40. As stated earlier, physical property change such as KV@40 is a consequence of compositional variation of individual components as well as their inter-molecular interactions. Therefore, the complex cross correlation pattern is not surprising.

The cross correlation pattern becomes simpler from 30 °C and simplest at 50 °C. Although the asynchronous 2D correlation maps acquired at other than 20 or 50 °C are not shown due to the simpler graphical comparison. From 50 to 90 °C, the corresponding correlations are more complex again. This result informs that, at 50 °C, individual molecular vibrations of components become less complex compared to those at other temperatures. So, the molecular vibrations and inter-molecular interactions featured at 50 °C derives the advent of the band at 975 cm^{-1} for the red-colored samples as shown in Fig. 4, and the use of selectivity-enhanced spectra are advantageous to describe the corresponding property change effectively.

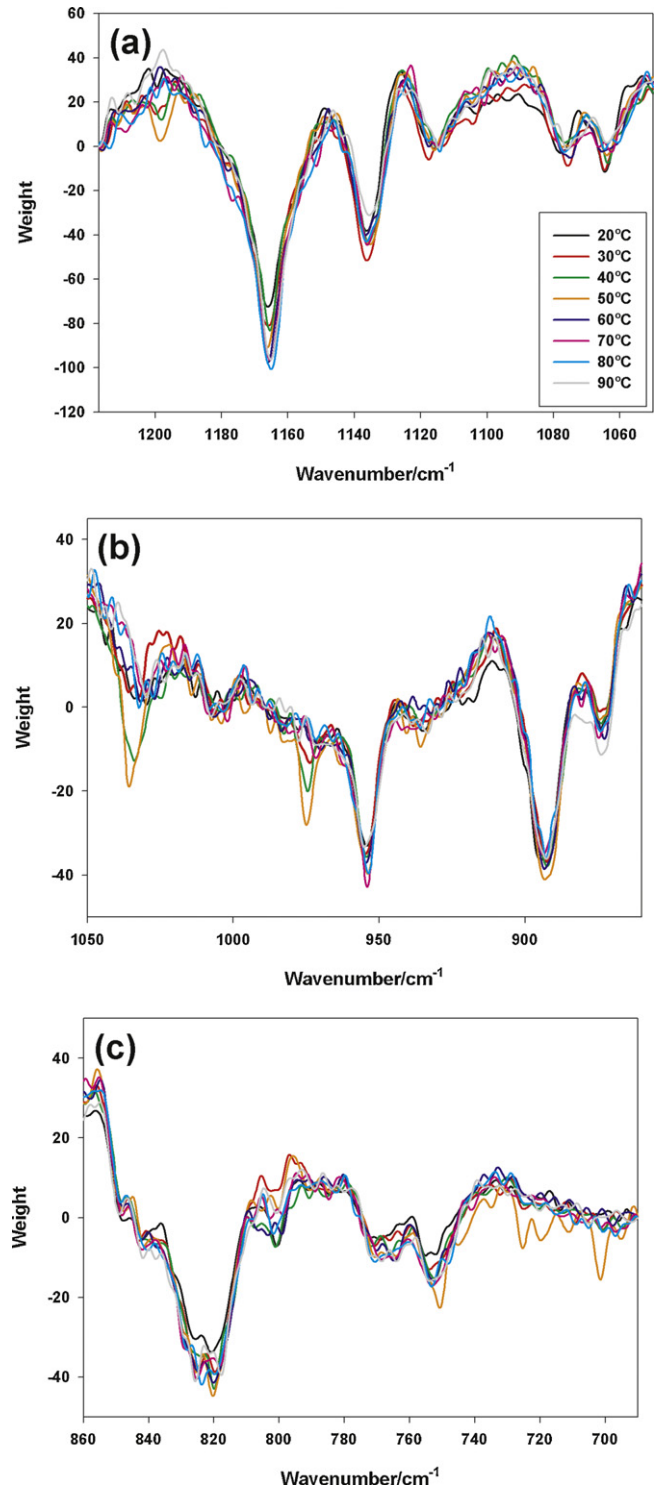

Fig. 5. The first PLS loadings at each measurement temperature in the 1217–1050 (a), 1050–860 (b) and 860–690 cm^{-1} (c) ranges. Each color indicates the temperature of measurement. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

3.6. Identification of adulterated olive oils at different measurement temperatures

We have also attempted the authentication of virgin olive oil using Raman spectroscopy as another example to demonstrate

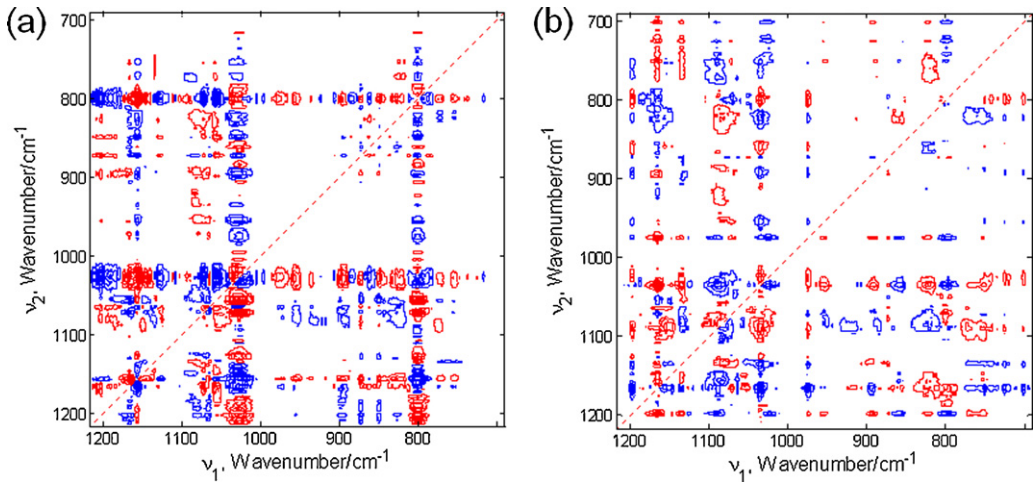

Fig. 6. Asynchronous 2D correlation maps obtained using the datasets collected at 20 (a) and 50 °C (b). The red and blue colors indicate the positive and negative cross peaks, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

analytical utility of the proposed strategy. For this study, 5% soybean oil-adulterated olive oils were tested for the discrimination against pure olive oils. The selection of 5% adulteration was to evaluate the performance of discrimination when the degree of adulteration is low. If olive oils are adulterated with high concentration of soybean oil, the discrimination would be distinct regardless of measurement temperatures. So, the use of highly adulterated samples is not proper in this study.

Fig. 7 shows normalized Raman spectra of pure olive (bottom) and soybean oil (top) collected at 8 different temperatures. The normalization procedure was the same as performed for LBO spectra. Each spectrum in the figure corresponds to average of 50 spectra acquired from 10 samples at each temperature. The spectra are offset for clear comparison. As shown, overall spectral features of olive and soybean oils are generally similar each other, and minor spectral differences are observed in the 1280–1250 and 910–790 cm^{-1} ranges. The intensity of 1265 cm^{-1} band, =C—H bending in *cis* RHC=CHR, is higher for soybean oils. As found in previous publications [19–23], the band intensity was useful to examine the degree of unsaturation. It increased as the degree of unsaturation became higher. In addition, Zhang et al. revealed that the intensity of 1265 cm^{-1} band was highly correlated with the content of unsaturated fatty acids in oils and increased with the higher soybean oil concentration [24]. So, the 1265 cm^{-1} band could be useful for identification of soybean oil-adulterated olive oils by soybean oils. The 1120 and 880 cm^{-1} bands with weak intensity are observable in the spectra of olive oils. On the other hand, these are nearly absent in the soybean oil spectra.

The spectral features of adulterated olive oils are quite similar to those of pure olive oils due to the only 5% contamination by soybean oils. Under this situation of low degree adulteration, the spectral discernability between two components is a critical factor governing the identification accuracy. Since the molecular structures of olive and soybean oils vary individually with the change of temperature, a certain temperature providing enhanced spectral discrimination between two components would possibly exist. The molecular structures of both oils are generally similar and the degree of adulteration is only 5%, so any analytical strategies capable of improving spectral discrimination should be greatly beneficial.

Using the spectral datasets collected at each temperature, the discrimination between pure and adulterated olive oils was performed using linear discriminant analysis (LDA). Initially PCA

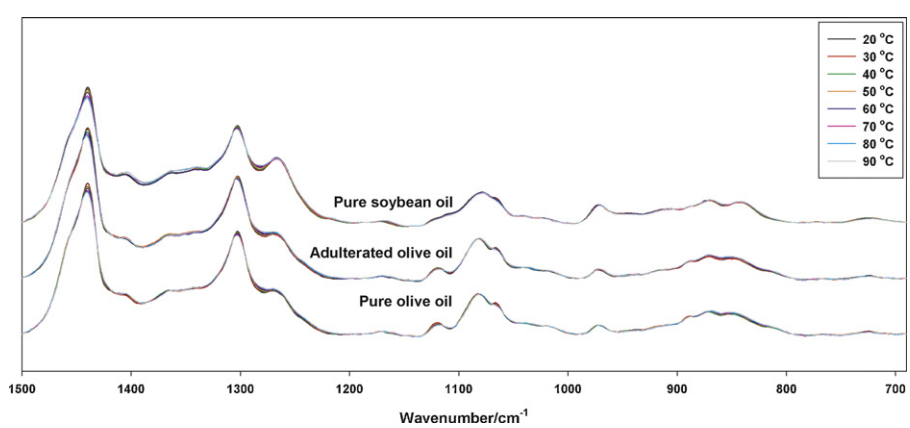

Fig. 7. Normalized Raman spectra (1500–690 cm^{-1} range) of pure olive oils (bottom), adulterated olive oils (middle) and pure soybean oils (top) collected at 8 different temperatures. Each spectrum in the figure corresponds to average of 50 spectra acquired from 10 samples at each temperature. The spectra are offset for clear comparison. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 1
Identification errors acquired at different temperatures using LDA combined with PCA.

Scores used	20 °C	30 °C	40 °C	50 °C	60 °C	70 °C	80 °C	90 °C
1st/2nd	28%	19%	22%	43%	45%	9%	48%	33%
1st/3rd	44%	45%	19%	8%	6%	35%	0%	0%
2nd/3rd	42%	16%	11%	6%	7%	6%	1%	8%

was performed by combining the spectra of pure and adulterated olive oils collected at each temperature, and the resulting scores were used for LDA. A combination of two scores was tested since the performance of discrimination was easy to visualize in the domain of two-dimensional score distribution. First to third scores obtained at each temperature were used for LDA.

Table 1 shows the results of LDA using the combinations of two separate scores for the discrimination between pure and adulterated olive oils. In the case of combining the first and second scores, the acquired discrimination errors are fairly high except the measurement at 70 °C. By contrast, when first and third scores are combined for LDA, the discrimination significantly improves especially at the 80 and 90 °C measurements, where two groups of pure and adulterated olive oils are perfectly discriminated. Based on the result, the third scores provide highly descriptive information for the discrimination. When the second and third scores are combined, the discrimination accuracy of 1% is achievable at the 80 °C measurement. In overall, the best discrimination was achieved by combining the first and third scores obtained from the measurements at either 80 or 90 °C.

Fig. 8(a) and (b) shows two-dimensional score scatter plots constructed using the first and third scores from the measurements at 20 and 90 °C, respectively. Blue and red circles indicate pure and adulterated olive oils, respectively. In each plot, the corresponding boundary line determined by LDA is also shown. Two groups are clearly distinct at the 90 °C measurement. By contrast, these are largely overlapped at the 20 °C measurement. It implies that the varied spectral features of olive and soybean oils around 80 and 90 °C aid the discrimination.

To explain the improved discrimination, the difference spectra acquired by subtracting adulterated olive oil spectrum from pure olive oil spectrum at the same temperature were initially examined as shown in Fig. 9(a). Each difference spectrum in the figure corresponds to average of 10 difference spectra acquired at each temperature. The 1500–1200 cm^{-1} range, where the spectral difference is relatively large, is shown. The difference spectra are arbitrarily offset for comparison. As shown, the largest and moderate spectral differences are observed around 1265 and 1432 cm^{-1}, respectively. The 1265 cm^{-1} band is related to the degree of unsaturation [19–24]. The difference spectral features at each temperature are dissimilar each other, indicating that characteristic spectral variations of olive and soybean oils at each temperature. The observation of the difference spectra could not directly provide explanation for the improved discrimination; however, the individually varied spectral features of olive and soybean oils should be the source for the improvement.

To further probe the origin of improved discrimination, the loadings used for PCA at each temperature were examined. Fig. 9(b) shows the first loadings in the 1450–1410 cm^{-1} range, highlighting 1432 cm^{-1} band. The weight on the 1432 cm^{-1} band is apparent for the measurements from 20 to 60 °C. Then, it shifts to the higher wavenumber at 70 °C and again moves back to 1432 cm^{-1} at 80 °C. At 90 °C, it suddenly disappears. Based on the examination, the abrupt spectral variation predominantly occurs from 70 to 90 °C and it is more dramatic between 80 and 90 °C. Fig. 9(c) and (d) shows the second and third loadings in the 1300–1200 cm^{-1} range, respectively, highlighting the 1265 cm^{-1} band. In the case of second loadings, the loading features also vary largely from 70 to 90 °C and the 90 °C-loading feature is substantially different compared with those of other temperatures. The third loading weight on the 1265 cm^{-1} band is weak for the measurements at 20 and 30 °C, and become greater up to 60 °C. Then, it decreases rather dramatically at 70 °C, and again increases at 80 and 90 °C. The shapes of third loadings at 80 and 90 °C are relatively clear with less wavy feature, possibly from noise, and the weighted band positions are slightly different each other. The variation of third loading weights according the temperature elevation is analogous to that of discrimination errors acquired with the combination of the first and third scores as shown in Table 1. When the weight on the 1265 cm^{-1} band is stronger and more distinct, the discrimination improves substantially. The varied spectral features of the 1265 cm^{-1} band, related to the degree of unsaturation [19–24], around 80–90 °C play an important role to enhance the discrimination between pure and adulterated olives.

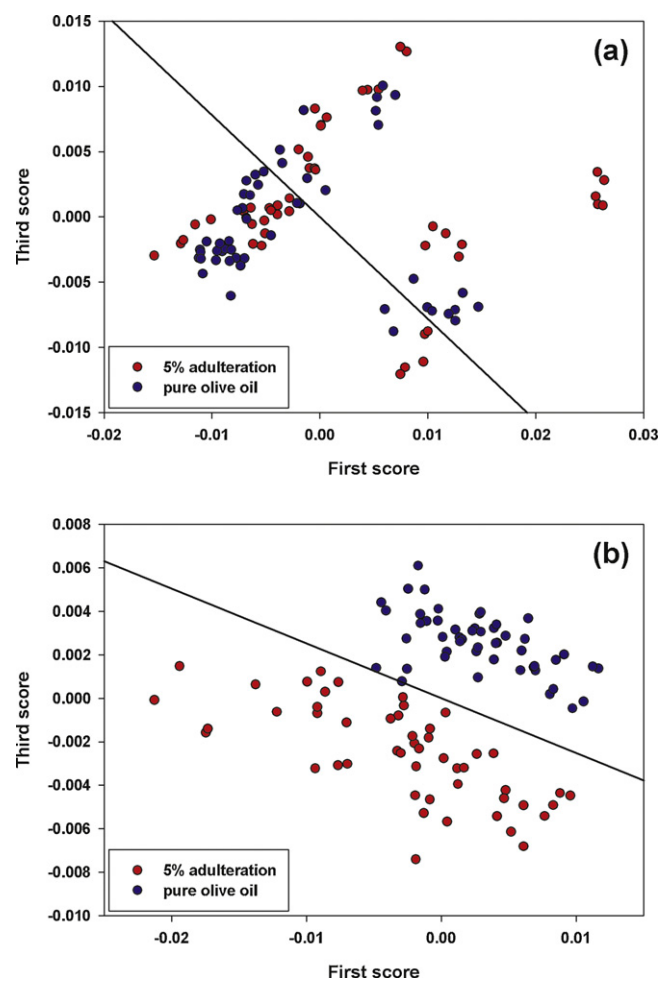

Fig. 8. Two-dimensional score scatter plots constructed using the first and third scores for the measurements at 20(a) and 90 °C (b). Blue and red circles indicate pure and adulterated olive oils, respectively. In each plot, the corresponding boundary line determined by LDA is also shown.

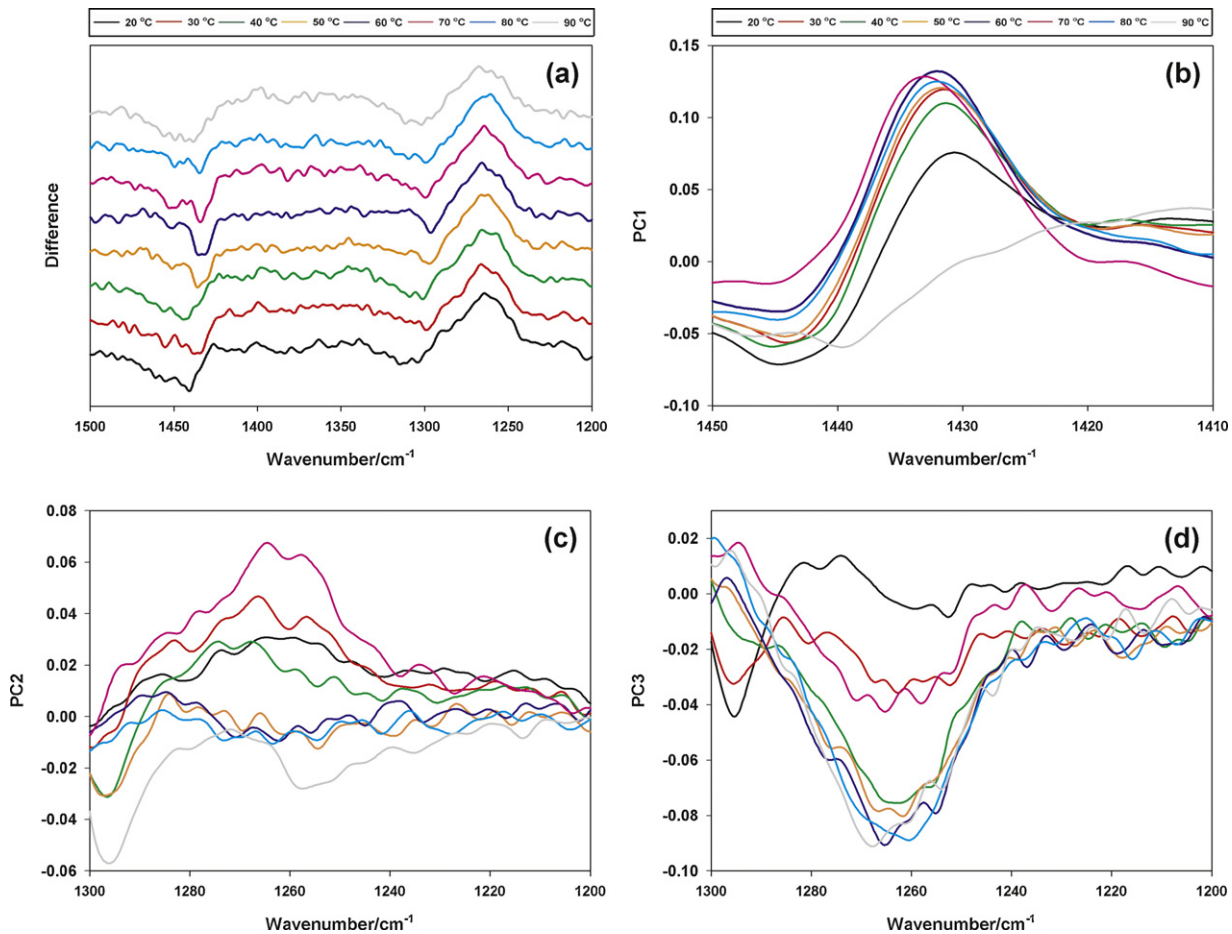

Fig. 9. Difference spectra acquired by subtracting adulterated oil spectrum from pure olive oil spectrum at the same temperature (a). Each difference spectrum in the figure corresponds to average of 10 difference spectra acquired at each temperature. The 1500–1200 cm^{-1} range, where the spectral difference is relatively large, is shown. The first loadings used for the measurements at each temperature in the 1450–1410 cm^{-1} range (b), and the corresponding second (c) and third loadings (d) in the 1300–1200 cm^{-1} range are shown, respectively.

4. Conclusion

The proposed strategy, utilizing temperature dependent Raman spectra to enhance the spectral features of samples, was effective to improve accuracy for determination of KV@40 for LBO samples and identification of adulterated olive oils. It is simple since only a device for temperature control of samples is required to execute the strategy. Our hope is to encourage analysts to probe a condition providing improved accuracy in a given spectroscopic analysis, rather than just measurement at room temperature. The method could be further applicable for analysis other complex samples such as petroleum-refining samples (e.g. gasoline, kerosene, diesel), complex mixtures of hydrocarbons with different structures and carbon numbers per molecule. Obviously, an optimized condition should vary depends on several parameters such as compositional complexity of samples and individual molecular structures of components. Although the results were based on the measurements in the 20–90 °C range, it would be worthwhile to expand the evaluation for measurements over 100 °C. In addition, theoretical and computational study to explain the temperature-induced spectral variation of individual components needs to accompany in future.

Acknowledgements

This research was supported by Basic Science Research Program through the National Research Foundation of Korea (NRF) funded by the Ministry of Education, Science and Technology (2012R1A1B3003965).

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